

Final Presentation

Rootkit Detection and System Hardening

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Overview/Outline

Outline

Goals,

Motivation,

Contributions

Background/Related Work

Work during Semester

Prior results

Design

Results

Expected Budget and Costs

Time Table

Conclusion/Summary

Demo

Goals

- Creating a Linux sandbox that allows me to evaluate its open-source tools against a compromised scenario of my choosing and making.
- Tools would include chkrootkit, rkhunter, and Lynis. The type of compromised scenario would be using an LD_PRELOAD
- Attempt system hardening to improve system security overall



Motivation



- Rootkits and malicious malware can be hard to detect
- Linux is widely used in cloud environments and runs the majority of web servers
- I wanted to gain more experience and build on anything I've known prior
- Allowed me to build a virtual sandbox without actually affecting my own system
- The project provided the opportunity to explore system hardening and security

Contribution

Established a Linux virtual environment

Successfully tested and managed open-source tools in the linux system

Deliverables- Security scans and analysis

Preload implemented in the environment

►Update- Ended up spending the amount I somewhat expect. Mostly thanks to the virtual machine environment.

Background/Related Work

What is a rootkit?

- ▶ A rootkit is a stealthy malicious software
- ▶ May bypass detection and goes unnoticed by the user and system.
- ▶ There are different types of kits that target different (*Avast, n.d*)

About tools I'm using...

Chkrootkit- scans for known signatures. ([Khandelwal](#), 2023)

Rkhunter- checks files and any suspicious changes

Lynis- for security auditing

Implementing a comprise/environment in the shape of an LD_PRELOAD

Building on the foundation learned in previous coursework.

What is system hardening?

Hardening is done after identifying weaknesses or vulnerabilities. The goal is to reduce, making the system more resistant.

Before Results during progress report

```
=====
Lynis security scan details:

Scan mode:
Normal [X] Forensics [ ] Integration [ ] Pentest [ ]

Lynis modules:
- Compliance status [?]
- Security audit [V]
- Vulnerability scan [V]

Details:
Hardening index : 65 [##### ]
Tests performed : 254
Plugins enabled : 1

Software components:
- Firewall [X]
- Intrusion software [X]
- Malware scanner [V]

Files:
- Test and debug information : /var/log/lynis.log
- Report data : /var/log/lynis-report.dat
=====
```

► Lynis security scan

- Original results before implementing anything else.
- Produced pretty normal results, scores are fairly average
- Wanted to see if we could get it higher by the end of the project
- what else could we find

Work for this Semester

- ▶ Linux research, rootkit concepts, compromised scenarios/ideas
- ▶ Building a Sandbox environment on Oracle VirtualBox along with setting the right specs.
- ▶ Installing and configuring open-source Linux security tools
- ▶ Performing scans and analyzing/comparing results
- ▶ Applying security hardening towards the end of my project
- ▶ Document finding in my reports



LD_PRELOAD (Simulated compromise idea)

Linux features that illustrates rootkit concepts, without modifying kernel.

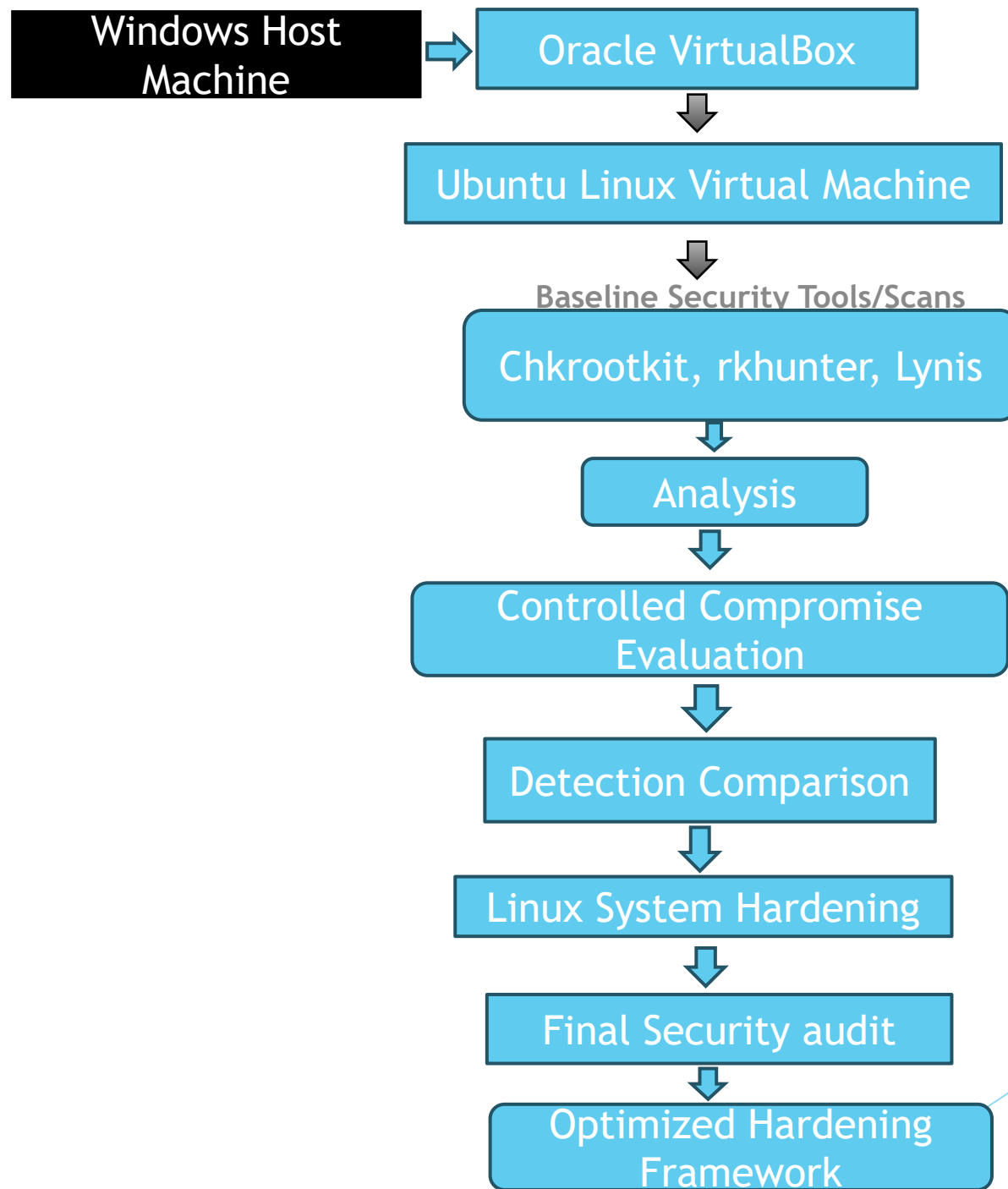
Works by allowing the program to load custom library, changing how programs behave

Build it, and establish it against the open-source tools

Source code c

Behaviors can change without changing original programs

Design



Budget/Cost

- ▶ VirtualBox download- Free
- ▶ Ubuntu Linux/Virtual Machine - Free
- ▶ Personal machine- Already paid for
- ▶ Tools and resources- Free
- ▶ Book- *Linux command line and shell scripting* - \$30. (Would help with ld_preload)
- ▶ Budget- A few hundred if needed \$\$\$
- ▶ Update- Ended up spending the amount I somewhat expect. Mostly thanks to the virtual machine environment.



Time Table

- ▶ 6 days out of the week
- ▶ Increased time on project to 2.5 hours on each given day
- ▶ This includes, research, reading, writing, implementation of the VM and settings
- ▶ Ended up implementing an extra section before my system hardening phase (id 23) and moved everything a couple days ahead.
- ▶ Extra section was to implement a type of compromised environment in the linux system.
- ▶ Did end up moving a few things ahead and around.

ID	Task Mode	Task Name	Duration	Start	Finish	Jun 7, '26							Jun 14, '26							Jun 21, '26							Jun 28, '26						
						M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	
19		-Run Chkrootkit	1 day	Fri 6/19/26	Fri 6/19/26																												
20		-Run rkhunter	1 day	Fri 6/19/26	Fri 6/19/26																												
21		Run lynis security audit	1 day	Fri 6/19/26	Fri 6/19/26																												
22		Document, report, and screenshot	1 day	Sat 6/20/26	Sat 6/20/26																												
23		6. System Hardening Phase	3 days	Sun 6/21/26	Tue 6/23/26																												
24		-Configure firewall settings	1 day	Sun 6/21/26	Sun 6/21/26																												
25		-Review permission and user settings	1 day	Mon 6/22/26	Mon 6/22/26																												
26		-Apply any recommendations from Lynis	1 day	Mon 6/22/26	Mon 6/22/26																												
27		- Re-test Sytem	1 day	Tue 6/23/26	Tue 6/23/26																												
28		7. Analysis Phase	4 days	Wed 6/24/26	Sat 6/27/26																												
29		-Review and compare tool	1 day	Wed 6/24/26	Wed 6/24/26																												
30		-Analyze any security findings	1 day	Thu 6/25/26	Thu 6/25/26																												
31		-Compare strengths, weaknesses, vulnerabilities	1 day	Fri 6/26/26	Fri 6/26/26																												
32		-Assess improvement	1 day	Sat 6/27/26	Sat 6/27/26																												

Conclusion

- I set out to build a linux sandbox where I can manage security tools against a somewhat compromised type of situation (Pre_load, not rootkit)
- Evaluated scans and tools.
- Installed a preload into linux system
- I did experience tedious struggles along the way, even through building or trying to test. Learning to identify the outputs I wanted

References

Avast. (2021, July 22). *What is a rootkit & how to remove it?*
<https://www.avast.com/c-rootkit>

Wiz. (n.d.). *Linux rootkits explained, Part 1: Dynamic linker hijacking.* <https://www.wiz.io/blog/linux-rootkits-explained-part-1-dynamic-linker-hijacking>



Demo

Will include me evaluating installed tools I use throughout the project. Will attempt to remove the preload to see if tools behave different